



RESEARCH OUR
RECORDS

VETERANS' SERVICE
RECORDS

EDUCATOR
RESOURCES

VISIT US

AMERICA'S FOUNDING
DOCUMENTS

Federal Records Management

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Symbol Key

Preferred Formats ● ● ●

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Computer Aided Design (CAD)

Computer aided design (CAD) – CAD formats are vector graphics files that rely on mathematical expressions to create multi-dimensional computer graphics intended for use in engineering and manufacturing design. CAD programs can generate representations and animations of two and three-dimensional surface projections of objects.

● ● ● Preferred Formats

Preferred
Formats

Format Version



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Extensible 3D (X3D)

ISO/IEC 19775-1:2008:
(<http://www.web3d.org/files/specifications/19775-1/V3.2/index.html>)

Standard for the Exchange of Product Model Data (STEP)

ISO 10303-21:2016:
(<https://www.iso.org/standard/63141.html>)

ASTM E57 3D file format (E57)

1.0
ASTM E2807-11(2019):
(<https://www.astm.org/e2807-11r19.html>)

OpenDocument Graphics Format (ODG)

All versions
ISO/IEC 26300-1:2015
Information technology — Open Document Format for Office Applications (OpenDocument) v1.2 — Part 1: OpenDocument Schema:
(<https://www.iso.org/standard/66363.html>)

Acceptable Formats

Acceptable Formats	Format Specifications
Portable Document Format/Engineering (PDF/E)	ISO 24517-1:2008 Document management — Engineering document format using PDF -- Part 1: Use of PDF 1.6 (PDF/E-1): (http://www.iso.org/iso/catalogue/catalogue_tc/catalogue_detail.htm?csnumber=42274)
Universal 3D (U3D)	Universal 3D (U3D) format family. ECMA-363, Editions 1-4: (https://www.ecma-international.org/wp-content/uploads/ECMA-363_4th_edition_june_2007.pdf)
Product Representation Compact (PRC)	ISO 14739-1:2014 Document management — 3D use of Product Representation Compact (PRC) format — Part 1: PRC 10001: (https://www.iso.org/standard/54948.html)

Digital Audio

The Digital audio category encompasses formats used to encode recorded sound as machine readable files by converting acoustic sound waves into digital signals. Digital audio formats are generally composed of both a wrapper format, usually the common name associated with the file extension, and an encoding method or codec.

General requirements for digital audio records:

- Digitize to standards appropriate for the accurate preservation of analog material (e.g., audio cassettes, records). Examples of appropriate methods and formats are available on the [Products and Services](#) page;
- Transfer digital audio at a minimum of 16 bits per sample;
- Transfer digital audio at a minimum sample rate of at least 44.1 kHz, as encouraged.



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Preferred Formats

Preferred Formats	Format Version	Codecs	Format Specifications
Broadcast Wave (BWF)	0, 1 & 2	Linear Pulse Code Modulated Audio (LPCM)	European Broadcast Union (EBU). Tech Specification of the Broadcast Wave Format (BWF) – Version 1: (http://web.archive.org/web/20091229093941/http://tech.ebu.ch/docs/tech/tech3285.pdf) Specification of the Broadcast Wave Format (BWF) - Version 2: (https://tech.ebu.ch/docs/tech/tech3285.pdf)
Free Lossless Audio Codec (FLAC)	1.21	FLAC	FLAC Format Specification version 1.21: (http://flac.sourceforge.net/format.html)

Acceptable Formats

Acceptable Formats	Format Version	Codecs	Format Specifications
Audio Interchange Format (AIFF)	1.3	Linear Pulse Code Modulated Audio (LPCM)	Audio Interchange File Format: "AIFF" A Standard for Sampled Sound Files Version 1.3 Apple Computer, Inc.: (https://www.mmsp.ece.mcgill.ca/Documents/AudioFormats/AIFF/AIFF.html)
MPEG Audio Layer III (MP3)		MP3enc, Lame	ISO/IEC-11172-3 Information technology – Coding of moving pictures and associated audio for digital storage media at up to about 1.5 Mbit/s – Part 3: Audio: (http://www.iso.org/iso/iso_catalogue/catalogue_tc/catalogue_detail.htm?csnumber=22412) ISO/IEC 13818-3:1995 Information technology – Generic coding of moving pictures and associated audio information – Part 3: Audio: (http://www.iso.org/iso/home/store/catalogue_ics/catalogue_detail_ics.htm?csnumber=26797)
Waveform Audio File Format (Wave)		Linear Pulse Code Modulated Audio (LPCM)	Multimedia Programming Interface and Data Specifications 1.0: (https://www.mmio.org/)

Digital Moving Images



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Digital moving images consist of bitmap digital images or “frames” displayed in rapid succession at a constant rate, giving the appearance of movement. This category includes two subcategories: digital cinema which encompasses digitized film; and digital video (including both video digitized from analogue sources and born digital video).

General requirements for digital moving image records:

- Agencies must digitize to standards appropriate for accurate preservation of the original video and audio components, when converting analog material. Examples of appropriate methods and formats are available on [NARA’s Digitization Services Products and Services](#) page; and
- For reformatted video, 8-bit is acceptable but 10-bit is preferred.

Digital Cinema

Preferred Formats

Preferred Formats	Format Version	Codecs	Format Specifications
Digital Moving Picture Exchange Bitmap (DPX)	2.0	Uncompressed	Society of Motion Picture Television Engineers. SMPTE ST 268:2014 (DPX Version 2.0): (https://my.smpte.org/s/product-details?id=a1BVR0000007Nen)

Digital Video

Preferred Formats

Preferred Formats	Format Version	Codecs	Format Specifications
AV1 Image File Format (AVIF)	V. 1.0 & V.1.1	AV1	AV1 Image File Format (AVIF) (https://aomediacodec.github.io/av1-avif/latest-approved.html#biblio-av1)

Acceptable Formats

Acceptable Formats	Format Versions	Codecs	Format Specifications
Audio Video Interleaved Format (AVI)		Uncompressed 4:2:2	Multimedia F (<a 505="" 623="" 900="" 983"="" data-label="Image" href="http://www.</td></tr></table></div><div data-bbox=">

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QuickTime File Format (MOV)	Uncompressed 4:2:2	Apple QuickTime File Format Specification (ISO/IEC 14496-14:2003): (https://developer.apple.com/library/mac/documentation/QuickTime/QTFF/QTFFPreface/qtffPreface.html#//apple_ref/doc/uid/TP40000939)
Windows Media Video 9 File Format (WMV)	VC-1	Advanced Systems Format (ASF) Specification Revision 01.20.03 Microsoft Corporation December 2004: (https://download.microsoft.com/download/7/9/0/790fecaa-f64a-4a5e-a430-0bccdab3f1b4/ASF_Specification.doc  ) Windows Media Video 9 encoder: (http://msdn.microsoft.com/en-us/library/windows/desktop/ff819505(v=vs.85).aspx )
MPEG 4	H.264	ISO/IEC 14496-10:2022 Information technology — Coding of audio-visual objects — Part 10: Advanced video coding: (https://www.iso.org/standard/83529.html )
MPEG-2 Video (MPEG2)		ISO/IEC 13818-2:2000 Information technology -- Generic coding of moving pictures and associated audio information: Video: (https://www.iso.org/standard/61152.html )
Material Exchange Format (MXF)	J2K-losslessly-compressed	ST 377-1:2011 Material Exchange Format (MXF) — File Format Specification: (http://standards.smpte.org/content/978-1-61482-517-3/st-377-1-2011/SEC1.abstract?sid=63bac43b-e0e1-40a3-8019-d379a103987e ) ISO/IEC 15444-1:2004Information technology -- JPEG 2000 image coding system: Core coding system: (https://www.iso.org/standard/78321.html )

Born-Digital Photographs

Born-digital photographs are raster images of natural, real-world scenes or subjects captured by digital cameras, including aerial photography.

NITF is acceptable only for born-digital photography. NITF should not be used for digitized photographs.

Additional special requirements for digital photographs are described in [36 CFR 1237.28](#).

Preferred Formats

Preferred Formats	Format Versions	Format Specifications
Tagged Image File Format (TIFF)	4, 5, & 6	TIFF Revision 6.0 Final — June 3 (https://www.adobe.io/content ,  )



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Acceptable Formats	Format Versions	Format Specifications
JPEG File Interchange Format (JFIF) with Joint Photographic Experts Group (JPEG) compression	1.02	ISO/IEC 10918-5 Information technology – Digital Compression and coding of continuous-tone still images: JPEG Interchange File Format: (http://www.iso.org/iso/home/store/catalogue_tc/catalogue_detail.htm?csnumber=54989 )
Digital Negative (DNG)	1.6.0.0	Adobe Digital Negative (DNG) Specification Version 1.6.0.0: (https://helpx.adobe.com/content/dam/help/en/photoshop/pdf/dng_spec_1_6_0_0.pdf  )
Portable Network Graphics (PNG)	1.2	ISO/IEC 15948:2004 Information technology -- Computer graphics and image processing -- Portable Network Graphics (PNG): Functional specification: (http://www.iso.org/iso/iso_catalogue/catalogue_tc/catalogue_detail.htm?csnumber=29581 )
Jpeg2000 (JP2)	JP2 – Part 1	ISO/IEC 15444-1:2004 Information technology – JPEG 2000 image coding system: Core coding system: (https://www.iso.org/standard/78321.html )
National Imagery Transmission Format (NITF)	2.0 and 2.1	MIL-STD-2500A--Department of Defense Interface Standard: https://earth-info.nga.mil/php/download.php?file=cib-2500a 

Digitized Paper and Photographic Prints

Digitized paper and photographic prints are digital records created according to the requirements in 36 CFR 1236 E—Digitizing Permanent Federal Records by converting paper or other paper-based formats using reflective techniques to a digital form that is of sufficient authenticity, reliability, usability, and integrity to serve in place of the source record.

The format and compression codec requirements are in § 1236.48-File format requirements. Specifically, the regulation requires agencies to:

(a) Encode, retain, and transfer digitized records in one of the following file formats, either uncompressed or using one of the specified compression codecs in the tables below.

(1) Agencies that combine multiple uncompressed TIFF images into PDF/A files using JPEG2000 compression must perform the quality inspection step specified in § 1236.48-5 for PDF/A files.

(2) When using JPEG 2000 visually lossless compression, agencies must ensure that the compression to apply, not to exceed 20:1, by performing a visual inspection for artifacts that obscure or alter the information content.

Optical Character Recognition



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NARA understands that the ability to embed OCR'd text in PDF records enhances access to the records. NARA will accept PDF records that have been OCR'd using processes that do not substitute generated or modified content for the original bit-mapped image.

While it will accept PDF records with uncorrected OCR'd text, NARA will not accept PDF records resulting from OCR processes that alter the visible content, degrade the quality of the original bit-mapped image, or replace the original bit-mapped image with OCR'd text.

NARA will not accept digitized records in PDF that have been saved with lossy compression to reduce file size (e.g., JPEG, JBIG2). Such processes degrade the quality of the original image and may make such images unsuitable for archival preservation.

Applications use different terms for similar output results. Some examples of terms indicating prohibited outputs include:

- “Searchable Image - Compact”
- “Editable text”
- “Editable text and images”
- “PDF Normal”
- “Formatted Text & Graphics”
- “TruePage”
- JBIG1 or JBIG2
- The use of “lossy” compression

Terms indicating acceptable output include, but are not limited to:

- Searchable Image - Exact
- Searchable Image Text (original image)
- Searchable Image - Best Quality

 Acceptable Formats for Digitized Permanent Paper Records

Acceptable Formats	Format Versions	Acceptable Compression Codecs	Format Specifications
Tagged Image File Format (TIFF)	6	Uncompressed, Deflate (ZIP)	TIFF Revision 6.0 Final — June 3, 1992 Adobe Systems Incorporated: (https://www.adobe.io/content/dam/udp/en/open/standards/tiff/TIFF6.pdf  )
JPEG2000 part 1 (ISO/IEC 15444-1:2019)		JPEG 2000 part 1 core coding system lossless compression. Agencies may use up to 20:1 visually lossless compression	ISO/IEC 15444-1:2004 Information technology – JPEG 2000 image coding system: Core coding system: (https://www.iso.org/standard/78321.html 
Portable network graphics (PNG)	1.2	Deflate (ZIP)	ISO/IEC image specification



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PDF/A	Select any version of PDF/A that meets project requirements.	Deflate (ZIP), JPEG 2000 part 1 core coding system lossless compression.	PDF/A - Electronic document file format for long-term preservation: (https://www.pdfa.org/resource/iso-19005-pdfa/)
	However, do not use the attachments feature in PDF/A-3 or PDF/A-4.	Agencies may use up to 20:1 visually lossless compression.	

Requirements for digitized permanent photographic print records include the following:

- The agency must encode, retain, and transfer digitized photographic print records in one of the following file formats, either uncompressed or with one of the compression codecs specified in the table below.
- For a series of predominantly textual records with interspersed photographic prints, use the formats in the table for paper records. For a series of predominantly printed photographs, including those with paper records interspersed, use the file formats in the table below for photographic print records.

Acceptable Formats for Digitized Permanent Photographic Print Records

Acceptable Formats	Format Versions	Acceptable Compression Codecs	Format Specifications
Tagged Image File Format (TIFF)	6	Uncompressed, Deflate (ZIP)	TIFF Revision 6.0 Final — June 3, 1992 Adobe Systems Incorporated: (https://www.adobe.io/content/dam/udp/en/open/standards/tiff/TIFF6.pdf)
JPEG2000 part 1 (ISO/IEC 15444-1:2019)		JPEG 2000 part 1 core coding system lossless compression. Agencies may use up to 20:1 visually lossless compression	(https://www.iso.org/standard/78321.html)
Portable network graphics (PNG)	1.2	Deflate (ZIP)	ISO/IEC 15948 image processing specification (http://www.iso.org/iso/catalogue_detail.htm?csnumber=29581)



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Born-Digital Posters

Born-Digital posters are posters created on a computer using graphics software. Posters are generally large in format and usually printed and displayed for advertising and publicizing purposes. Paper posters must be digitized according to the requirements in 36 CFR 1236 E.

Preferred Formats

Preferred Formats	Format Versions	Format Specifications
Tagged Image File Format (TIFF)	4, 5 & 6	TIFF Revision 6.0 Final — June 3, 1992 Adobe Systems Incorporated: (https://www.adobe.io/content/dam/udp/en/open/standards/tiff/TIFF6.pdf)
Jpeg2000 (JP2)	Part 1 (JP2)	ISO/IEC 15444-1:2004 Information technology – JPEG 2000 image coding system: Core coding system: (https://www.iso.org/standard/78321.html)
Portable Network Graphics (PNG)	1.2	ISO/IEC 15948:2004 Information technology -- Computer graphics and image processing -- Portable Network Graphics (PNG): Functional specification: (http://www.iso.org/iso/iso_catalogue/catalogue_tc/catalogue_detail.htm?csnumber=29581)
Portable Document Format/Archival (PDF/A)	PDF/A-1	ISO 19005-1:2005 Electronic document file format for long-term preservation – Part 1: Use of PDF 1.4 (PDF/A-1): (http://www.iso.org/iso/catalogue_detail?csnumber=38920)
OpenDocument Graphics Format (ODG)	All versions	ISO/IEC 26300-1:2015 Information technology — Open Document Format for Office Applications (OpenDocument) v1.2 — Part 1: OpenDocument Schema: (https://www.iso.org/standard/66363.html)

Acceptable Formats

Acceptable Formats	Format Versions	Format Specifications
JPEG File Interchange Format (JFIF) with Joint Photographic Experts Group (JPEG) compression	1.02	ISO/IEC 10918-5 Information technology – Digital Compression and coding of continuous-tone still images: JPEG Interchange File Format: (http://www.iso.org/iso/home/store/catalogue_tc/catalogue_detail.htm?csnumber=54989)
Graphics Interchange Format (GIF)	87a & 89a	ISO/IEC 10918-1:1994 Information technology – Digital Compression and coding of continuous-tone still images: Requirements and guidelines: (http://www.w3.org/Graphics)



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Geospatial Formats

Geospatial records include digital cartographic data files and aerial photography that are created and processed in Geographic Information Systems (GIS) or other software applications for spatial analysis.

Preferred Formats

Preferred Formats	Format Versions	Format Specifications
Geospatial Tagged Image File Format	1.8.2	Geo TIFF Format Specification: (http://geotiff.maptools.org/spec/geotiffhome.html)
Geographic Markup Language	2.0 through 3.2	ISO 19136:2007 & Version 3.2, OGC document 07-036: (http://www.opengeospatial.org/standards/is)
Topologically Integrated Geographic Encoding and Referencing Files	2006 Second Edition	2006 Second Edition TIGER/Line®: (https://www.census.gov/programs-surveys/geography/technical-documentation/complete-technical-documentation.html)
Keyhole Markup Language	2.2	Open Geospatial Consortium OGC 07-147r2: (http://www.opengeospatial.org/standards/kml/)
Geopackage Encoding Standard	1.3.1	Open Geospatial Consortium OGC 12-128r18: (https://www.geopackage.org/spec131/index.html)

Acceptable Formats

Acceptable Formats	Format Versions	Format Specifications
Vector Product Format		MIL-STD-2407: (https://apps.dtic.mil/sti/pdfs/AD1110765.pdf)
ESRI ARC/INFO Interchange File Format		Reverse engineered specification: (http://avce00.maptools.org/docs/v7_e00_cover.html)
TerraGo Geospatial PDF	GeoPDF Encoding Best PracticeVersion 2.2	Open Geospatial Consortium Inc. OGC 08-139r2: (http://www.opengeospatial.org/standards/geopdf)
ESRI Shapefile (Compound)	1997 – current version	ESRI Shapefile Technical Specification: (http://www.esri.com/arcswt/shapefile/shapefile.htm)

Acceptable for Imminent Transfer Formats



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Acceptable for
Imminent
Transfer
Formats

Format Versions Format Specifications

Spatial Data
Transfer
Standard (SDTS)

All versions

ANSI NCITS 320-1998:
(<https://pubs.usgs.gov/fs/1999/0077/report.pdf> )

Presentation Formats

Presentation formats are used to convey graphical information to audiences in the form of a slide show. Presentation formats are not acceptable for use as transfer containers for permanent digital still images.

 Preferred Formats

Preferred
Formats

Format Versions Format Specifications

OpenDocument
Presentation
Format (ODP)

All versions

ISO/IEC 26300-1:2015
Information technology — Open Document Format for
Office Applications (OpenDocument) v1.2 — Part 1:
OpenDocument Schema:
(<https://www.iso.org/standard/66363.html> )

Portable
Document
Format Archival
(PDF/A-1)

PDF/A-1

ISO 19005-1:2005 Electronic document file format for
long-term preservation
– Part 1: Use of PDF 1.4 (PDF/A-1):
([http://www.iso.org/iso/catalogue_detail?
csnumber=38920](http://www.iso.org/iso/catalogue_detail?csnumber=38920) )

 Acceptable Formats

Acceptable
Formats

Format
Versions Format Specifications

Microsoft
Powerpoint
1997-2007
Binary
Format
(PPT)

8.0

[MS-PPT]: PowerPoint (.ppt) Binary File Format:
([http://msdn.microsoft.com/en-us/library/cc313106\(v=office.12\).aspx](http://msdn.microsoft.com/en-us/library/cc313106(v=office.12).aspx) )

Microsoft
Powerpoint
Office
Open XML
Format
(PPTX)

[MS-OI29500]: Office Implementation
Standards Support:
([http://msdn.microsoft.com/en-us/library/cc313106\(v=office.12\).aspx](http://msdn.microsoft.com/en-us/library/cc313106(v=office.12).aspx) )



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PDF/A-2 PDF/A-2 ISO 19005-2:2011 Document management -- Electronic document file format for long-term preservation -- Part 2: Use of ISO 32000-1 (PDF/A-2): (http://www.iso.org/iso/home/store/catalogue_tc/catalogue_detail.htm?csnumber=50655)

Born-Digital Textual Data

The born-digital textual data category refers to two general content types: unformatted (plain text) or formatted. Unformatted plain text (defined in MIME as text/plain) contains basic character information and control or non-printing characters but lacks styling information. Formatted text files include all of the attributes of plain text files but have extended formatting capabilities, for “stylized” or “rich” text features including italics, bold, colors, hyper-linking, etc.

Agencies must identify the character encoding method used with each text file.

Preferred Formats

Preferred Formats	Format Versions	Format Specifications
ASCII Text	7 bit	ISO/IEC 646:1991 Information technology -- ISO 7-bit coded character set for information interchange: (http://www.iso.org/iso/catalogue_detail.htm?csnumber=4777)
Unicode Text	UTF-8	RFC 3629: UTF-8, A Transformation Format of ISO 10646: (http://tools.ietf.org/html/rfc3629)
	UTF-16	RFC 2781 UTF-16: An Encoding of ISO 10646: (http://www.ietf.org/rfc/rfc2781.txt)
OpenDocument Text Format (ODT)	All versions	ISO/IEC 26300-1:2015 Information technology — Open Document Format for Office Applications (OpenDocument) v1.2 — Part 1: OpenDocument Schema: (https://www.iso.org/standard/66363.html)
PDF/A-1	PDF/A-1	ISO 19005-1:2005 Document management -- Electronic document file format for long-term preservation -- Part 1: Use of PDF 1.4 (PDF/A-1): (http://www.iso.org/iso/catalogue_detail?csnumber=38920)
PDF/A-2	PDF/A-2	ISO 19005-2:2011 Document management -- Electronic document file format for long-term preservation -- Part 2: Use of ISO 32000-1 (PDF/A-2): (http://www.iso.org/iso/home/store/catalogue_tc/catalogue_detail.htm?csnumber=50655)

Acceptable Formats

Acceptable Formats	Format Versions	Format Specifications
PDF	PDF 1.7	ISO 32000-1:2008 document format for long-term preservation



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PDF 1.0-1.6

(http://www.iso.org/iso/catalogue_detail.htm?csnumber=51502)

Adobe® Portable Document Format Version 1.6:
(<https://opensource.adobe.com/dc-acrobat-sdk-docs/pdfstandards/pdfreference1.6.pdf>)

Microsoft Word
(DOCX) Office
Open XML

OOXML Microsoft
Word for
Windows, version
2007-2010

[MS-OI29500]: Office Implementation Information for
ISO/IEC 29500 Standards Support:
(<http://msdn.microsoft.com/en-us/library/ee908652%28v=office.12%29>)

Microsoft Word 97
Binary Document
Format (DOC)

8.0

[MS-DOC]: Word (.doc) Binary File Format:
(<http://msdn.microsoft.com/en-us/library/cc313153%28v=office.12%29.aspx>)

Case Files

A case file is a collection of records that relate to a specific matter, individual, event, or administrative action. For example, documents related to [court cases](#) or other investigations or any other logical groupings of records in different file formats.

As they facilitate distribution of potentially large and diverse sets of content, case files elevate the risk that personally identifiable information (PII) or other sensitive information could inadvertently be made public if contained in a case file.

General Requirements for Case Files

1. Case files must relate to one another as components of a record. For example, a case file might contain the files that make up a medical case file or a court proceeding.
2. All files must be in a format identified, on this page, in NARA Bulletin 2014-04 Appendix A, as being either preferred or acceptable for use for their record type.
3. All data must be plainly viewable and not hidden. Many formats have the ability to hide sections of data, for example, a hidden column in a Microsoft Excel spreadsheet.
4. Agencies must deactivate any security settings (for example, self-sign security, user passwords, and/or permissions) on the case file that prevent NARA from opening, viewing, or printing the record

Metadata Fields for Embedded Files in Case Files:

1. In addition to the CSV file required by [NARA Bulletin 2015-04](#), PDF collections must be accompanied by an external index listing all embedded files. The index must be either in a Microsoft Excel or CSV file. The index file should also be contained within the collection.
2. Indexes must have the same name as the files in the case file they describe. Example: Case-File-001.pdf and Case-File-001.csv.

Index of Required Metadata Fields:

Metadata Element	Label	Definition
File Name	Identifier:FileName	The complete name of the embedded file, including the file extension. File names must be unique and include the file content. Example: Depo



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		001.docx. [Category-case file number-sequential number.file_extension]	
Message Digest	MessageDigest	The MD5 Checksum value of the embedded file	Mandatory
Title	Title	The descriptive name given to the embedded file. For example: Docket, Notes, Correspondence, Evidence	Optional. Mandatory if available
Creation Date	CreationDate	The date the embedded file was originally created	Mandatory
Modified Date	ModifiedDate	The date the embedded file was last changed	Mandatory
Access Restrictions	AccessRestrictions	Any PII, CUI, or National Security Restrictions that apply to the embedded file. If no restrictions apply label: No Access Restrictions	Mandatory

PDF Collections

PDF collections, commonly known as PDF Portfolios or containers, are a special type of PDF file that allows embedding additional files. Collections provide a useful way to package together case files.

Embedded files are stored in their original formats, and are displayed in the PDF in a manner similar to that of file system explorers or zip containers. PDF collections are also conventional PDF files and can include pages of text, scanned text, forms, or any other features supported by the PDF format.

Files in any other format, including other PDF files, may be embedded in PDF collections although the software used to create or view them may prohibit embedding or extracting certain file types, such as executable files, for security reasons.

Unless directly supported by the PDF viewing software, an embedded file must be extracted from the PDF and opened by an application capable of rendering its file format.

General Requirements:

1. Agencies may include any number of email or text messages that are components of a case file as embedded files. However, PDF collections must not be used to transfer aggregations of email records under a Capstone schedule. Acceptable formats for use when transferring email records are listed in the Email section of this table.
2. All PDF files including both the PDF collection itself and any embedded PDF file must have all fonts, including the [base 14 fonts](#), embedded within them. This requirement is met by embedding subsets of all fonts used in the document, or by saving as a PDF/A-3 or PDF/A-4f file.
3. Embedded files in PDF Collections must be visible. For example, pages in PDF Collections should not contain hidden files such as file attachment annotations.
4. PDF collections must not use folders to store or organize embedded files.
5. Embedded files should not contain additional embedded files. If a PDF collection contains a total of two hierarchical levels, the PDF collection itself counts as the first level.

			Preferred Formats
Preferred Formats	Format Versions	Format Specifications	



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PDF/A PDF/A-4f PDF format: (<https://pdfa.org/pdf-a-4/>)

PDF/A PDF/A-3 PDF format: (<https://pdfa.org/pdf-a-3/>)

Acceptable Formats

Acceptable Formats	Format Versions	Format Specifications
PDF	PDF 2.0	PDF format: (https://www.pdfa.org/pdf20)
PDF	PDF 1.7	ISO 32000-1:2008 Document management -- Portable document format -- Part 1: PDF 1.7: (http://www.iso.org/iso/catalogue_detail.htm?csnumber=51502) or (https://pdfa.org/iso-32000-1/)

Structured Data Formats

Structured data comprises the broad category of data that is stored in defined fields. Categories for structured data are as follows:

- Database formats are organized collections of associated data that conform to a logical structure. Database formats are determined by “data models” that describe specific data structures used to model an application and generally include navigational, relational, and hybrid models;
- Spreadsheets are tables made up of columns and rows and which contain cells of data. Relationships between cells can be pre-defined as mathematical formulas;
- Statistical data is the result of quantitative research and analysis. Statistical data formats contain collections of data presented in both tabular and non-tabular form; and
- Scientific data refers to research data collected by instrumentation tools during the scientific process. Scientific data formats are either domain specific within a single field of study, or are multi-domain formats used for transfer of scientific data between domains.

General requirements for structured data include the following:

- Agencies must transfer structured data that is both well-formed according to the syntactical conventions of the format, and valid according to the structural rules defined in any associated schemas or document type definitions (DTDs);
- Value Separated Files, e.g. CSV or comma separated value files, may use a character other than the comma. The pipe or caret are recommended delimiters because they are not commonly found in free text fields. Alternatively, text files encoded with ASCII characters and where each field is a fixed width, is also an acceptable transfer format for use with structured data, even though ASCII is technically a data encoding type. ASCII text files must be accompanied by complete documentation of the record lengths and field widths;
- Data files and databases shall be transferred as flat files or as rectangular tables, that is, as two-dimensional arrays, lists or tables. All records in a database, or rows (tuples) in a relational database, should have the same logical format. Each data element within a record should contain only one data value. A record should not contain nested repetitions;
- Structured data must be transferred together with any information that is necessary to ensure the validity of the data, e.g., DTDs, schemas, and data dictionaries.

Preferred Formats

Preferred Formats	Format Versions	Format Specifications
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Comma Separated Value (CSV)	N/A	Common Format and MIME Type for Comma-Separated Values (CSV) Files: (http://tools.ietf.org/html/rfc4180)
OpenDocument Format Spreadsheet (ODS)	All versions	ISO/IEC 26300-1:2015 Information technology — Open Document Format for Office Applications (OpenDocument) v1.2 — Part 1: OpenDocument Schema: (https://www.iso.org/standard/66363.html)
ASCII Text	7 bit	ISO/IEC 646:1991 Information technology -- ISO 7-bit coded character set for information interchange: (http://www.iso.org/iso/catalogue_detail.htm?csnumber=4777)
JavaScript Object Notation (JSON)		The application/json Media Type for JavaScript Object Notation (JSON): (http://www.ietf.org/rfc/rfc4627.txt?number=4627)
Extensible Markup Language (XML)	1.1	Extensible Markup Language (XML) 1.1 (Second Edition): (http://www.w3.org/TR/2006/REC-xml11-20060816/)
Software Independent Archiving of Relational Databases (SIARD)		SIARD Format Specification: (https://siard.dilcis.eu/specification/SIARD_2_2.pdf)

Acceptable Formats

Acceptable Formats	Format Versions	Format Specifications
Microsoft Excel Office Open XML	OOXML Workbook Excel 2007-2010 XLSX Microsoft Excel for Windows, version 2007	[MS-OI29500]: Office Implementation Information for ISO/IEC 29500 Standards Support: (http://msdn.microsoft.com/en-us/library/ee908652%28v=office.12%29)
Microsoft Excel 97 Binary Document Format (XLS)	Version 8.0	[MS-XLS]: Excel Binary File Format (.xls) Structure: (http://msdn.microsoft.com/en-us/library/cc313154(v=office.12).aspx)

Acceptable for Imminent Transfer Formats

Acceptable for Imminent Transfer Formats	Format Versions	Format Specifications
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Extended Binary
Coded Decimal
Interchange Code
(EBCDIC)

U.S. EBCDIC

IBM EBCDIC:
(<https://en.wikipedia.org/wiki/EBCDIC>)

Email

Email is defined as discrete electronic communications transmitted over the Simple Mail Transfer Protocol (SMTP), between two or more people or entities, in compliance with applicable IETF’s Request for Comments (RFC) specifications. Email does not include other functions commonly available via email programs such as calendars, tasks, appointments, newsgroups, or instant messaging. In order for information in a calendar, contact list, address book etc. to be transferred to NARA, it must be scheduled as a separate item.

Please note that NARA considers email attachments to be a component of the email record and does not require that unseparated email attachments meet the transfer standards specified by the format category under which the attachment alone would fall.

General requirements for email:

- Transfers of email records must consist of an identifiable, organized body of records (not necessarily a traditional series);
- Email messages should include delimiters that indicate the beginning and end of each message and the beginning and end of each attachment, if any. Each attachment must be differentiated from the body of the message, and uniquely identified;
- Email messages transferred as XML files must be accompanied by any associated document type definitions (dtds), schemas, and/or data dictionaries;
- Labels to identify each part of the message (Date, To [all recipients, including cc: and bc: copies], From, Subject, Body, and Attachment) including transmission and receipt information (Time Sent, Time Opened, Message Size, File Name, and similar information, if available). To ensure identification of the sender and addressee(s), agencies that use an email system that identifies users by codes or nicknames, or identifies addressees only by the name of a distribution list should include information with the transfer-level documentation; and
- Email converted to formats not natively used by the email program, and which do not maintain header information (such as RTF or Word documents) are not accepted. Printouts of emails are also not accepted under this Bulletin.

Preferred Formats for Individual Messages

Preferred
Formats for
Individual
Messages

Format Versions

Format Specifications

Internet Message Format:
(<http://www.ietf.org/rfc/rfc2822.txt>)

Internet Message
Format
(EML)

And MIME:
(<http://tools.ietf.org/html/rfc2045>)
(<http://tools.ietf.org/html/rfc2046>)
(<http://tools.ietf.org/html/rfc2047>)
(<http://tools.ietf.org/html/rfc2048>)
(<http://tools.ietf.org/html/rfc2049>)



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MBOX Email Format:
(<https://tools.ietf.org/html/rfc4155>)

MBOX Email
Format (MBOX)

And MIME:
(<http://tools.ietf.org/html/rfc2045>), (
<http://tools.ietf.org/html/rfc2046>), (
<http://tools.ietf.org/html/rfc2047>), (
<http://tools.ietf.org/html/rfc4288>), (
<http://tools.ietf.org/html/rfc4289>), (
<http://tools.ietf.org/html/rfc2049>)

Acceptable Formats for Individual Messages

Acceptable Formats for Individual Messages

Format Versions

Format Specifications

Extensible
Markup Language
(XML)

1.1

Extensible Markup Language (XML) 1.1 (Second
Edition):
(<http://www.w3.org/TR/2006/REC-xml11-20060816/>)

Microsoft Outlook
Item Message
Format (MSG)

Microsoft Outlook Item Message Format:
([http://msdn.microsoft.com/en-us/library/cc463912\(v=exchg.80\).aspx](http://msdn.microsoft.com/en-us/library/cc463912(v=exchg.80).aspx))

Preferred Formats for Aggregations of Email

Preferred Formats for Aggregations of Email

Format Versions

Format Specifications

Microsoft
Personal Folders
Format (PST)

2003 (Unicode)

Outlook Personal Folders File Format 2003 (Unicode):
(https://learn.microsoft.com/en-us/openspecs/office_file_formats/ms-pst/141923d5-15ab-4ef1-a524-6dce75aae546?redirectedfrom=MSDN)

MBOX Email
Format (MBOX)

MBOX Email Format: (
<https://tools.ietf.org/html/rfc4155>)

And MIME:
(<http://tools.ietf.org/html/rfc2045>), (
<http://tools.ietf.org/html/rfc2046>), (
<http://tools.ietf.org/html/rfc2047>), (
<http://tools.ietf.org/html/rfc4288>), (
<http://tools.ietf.org/html/rfc4289>), (
<http://tools.ietf.org/html/rfc2049>)



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Acceptable for
Imminent
Transfer
Formats

Format Versions Format Specifications

Microsoft
Personal Folders
Format (PST)

1997-2002 (ANSI)

Outlook Personal Folders File Format 1997-2002 (ANSI):

(https://learn.microsoft.com/en-us/openspecs/office_file_formats/ms-pst/141923d5-15ab-4ef1-a524-6dce75aae546?redirectedfrom=MSDN)
[↗](#)

Web Records

Web records consist of web sites and social media sites created and maintained to provide information and services of the United States Government via the World Wide Web. This Bulletin applies to web records managed by an agency that have been appraised and scheduled for permanent retention by NARA. Agencies should harvest websites using a utility that will package component files in a manner that meets the following general requirements.

General requirements for web content records:

- Web records must be accessible via Hypertext Transfer Protocol (HTTP) from a server to a client *browser* when a *URL* has been activated;
- Web content records that share a domain name including content managed under formal agreement and residing on another site must be transferred together;
- All component parts of web content records that have been appraised as permanent including image, audio, video and all other proprietary formats, must be transferred in a manner that maintains all of the original links, functionality and data integrity;
- Dynamic content such as calendars or databases either must be transferred in an acceptable format, or be made accessible as static content;
- All internally referenced URLs must be included with the transfer set; and
- All control information from the harvesting protocol must be maintained.

The following will not be accepted for transfer under this Bulletin:

- Program or administrative records documenting the management of web sites;
- Externally referenced content (e.g., accessed via hyperlink) that resides in a different domain and is not managed for an agency under a formal agreement;
- Static images, (such as screen shots), of web content records, because they do not retain hypertext functionality.

● ● ● Preferred Formats

Preferred
Formats

Format Versions

Format Specifications

Web ARChive
Format (WARC)

1.0

WARC file format
(<https://iipc.gov/specifications/1.0/index.html>)



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Web ARChive Format (WARC)	1.1	WARC file format: (https://iipc.github.io/warc-specifications/specifications/warc-format/warc-1.1/index.html)
Web Archive Collection Zipped (WACZ)	All versions	WACZ file format: (https://specs.webrecorder.net/wacz/1.1.1/)

Acceptable Formats

Acceptable Formats	Format Versions	Format Specifications
Archive File Format (ARC)	1.0	Arc File Format: (http://archive.org/web/researcher/ArcFileFormat.php)

Calendars

Electronic calendars allow users or groups to create events that can be exchanged between applications or systems. Users can manage and view events across defined time periods, such as hours, days, weeks, months, and years. The format, iCalendar or iCal, contains a variety of functions: events, to-do lists, journal entries, time zone, availability designated as free or busy, and notifications. The format can be exchanged through a variety of methods including Simple Mail Transfer Protocol (SMTP), HyperText Transfer Protocol (HTTP), a file system, and protocols, such as memory-based clipboard or drag and drop functions.

Calendar events can include attachments in multiple file formats, such as documents or spreadsheets or other file formats not specified within this appendix. Attachments are not required to be transferred separately. As components of the calendar record, attachments are transferred within the calendar file.

General requirements for calendar files:

- Transfers of calendar records must consist of an identifiable, organized body of records (not necessarily a traditional series);
- Calendars must include labels to identify each component. Delimiters must indicate the beginning and end of each event and the beginning and end of each attachment, if any. Each attachment must be differentiated from the body of the calendar. Each attachment must be uniquely identified; and
- NARA will not accept transfers of calendars converted to PDF files.

Preferred Formats

Preferred Formats	Format Versions	Format Specifications
iCalendar / iCal (ICS)	N/A	Internet Calen Specification (Engineering Ta (https://www.i

Navigational Charts



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Electronic navigational charts are tools used to provide safe navigation for all classes of vessels. Official charts, issued by a government, hydrographic office, or other government institution, comply with standards developed by the International Hydrographic Organization (IHO). These charts show detailed configurations of the seabed, characteristics of the coast, routes, and aids to navigation.

There are two types of electronic navigational charts—raster charts and vector charts:

- Electronic Navigational Charts (ENC) are vector data sets of all the objects (points, lines, and areas) represented on a chart in a digital database.
- Raster Navigational Charts (RNC) are digitized raster copies of official paper charts.

 Preferred Formats

Preferred Formats	Format Versions	Format Specifications
Electronic Navigational Chart	S/57	IHO Transfer Standard for Digital Hydrographic Data, Special Publication No. 57 (https://iho.int/uploads/user/pubs/standards/s-57/31Main.pdf  )
Raster Navigational Chart	BSB	Product Specification for Raster Navigational Charts (RNC), Special Publication No. 61 (https://iho.int/uploads/user/pubs/standards/s-61/S61E.pdf  )

Seismic Data

Seismic data formats capture data traces of seismic responses at specific surface recording locations. Transfers of seismic data in the SEG-Y format should include all associated metadata and support documentation needed to comprehensively understand the record. Additionally, metadata documenting the process of creating the data (if available) should be included. Examples of associated files and support documentation that should be included with transfers include: Load Sheets, Observer’s Reports, Readme files, and Velocity Files.

 Preferred Formats

Preferred Formats	Format Versions	Format Specifications
SEG-Y Data Exchange	Rev. 2.1	SEG-Y format: (https://library.seg.org/pb-assets/technical-standards/seg_y_rev2_1-oct2023-1701361639333.pdf  )

 Acceptable Formats

Acceptable Formats	Format Versions	Format Specifications
SEG-Y Data Exchange	Rev. 1.0	SEG-Y Format: (https://librarystandards/seg



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Updated August 2025



PDF files require the [free Adobe Reader](#).
More information on Adobe Acrobat PDF files is available on our [Accessibility page](#).

 This page was last reviewed on March 11, 2026.
[Contact us with questions or comments.](#)

Records Management

Contact Us

- Federal Records Officers
- NARA Appraisal Archivists
- Senior Agency Officials List

Email Us

- Office of the Chief Records Officer

Our Blog

- Records Express

I Want to Know More About...

- Accessioning Records
- Basic Laws & Authorities
- Email Management
- BRIDG Meetings
- Essential Records and Disaster Recovery

Records Management Resources

- Code of Federal Regulations
- FAQs
- Memos to Agency Records Officers
- Publications
- Records Management Glossary of Terms
- Records Storage Facility Standards Toolkit
- RM Consulting Services
- Scheduling Tools
- FRC Toolkit

Work Groups

- Appraisal
- General Records Schedule (GRS)
- Inspections and Oversight
- Records Control Schedules (RCS)
- Training





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